



**CALICUT UNIVERSITY – FOUR-YEAR UNDER
GRADUATE PROGRAMME (CU-FYUGP)**

BSc PHYSICS HONOURS

Programme	B.Sc. Physics Honours				
Course Title	NONLINEAR DYNAMICS AND CHAOS				
Type of Course	Major Elective				
Semester	VIII				
Academic Level	400 - 499				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	4	-	-	60
Pre-requisites	Numerical Techniques, Classical Mechanics				
Course Summary	To understand the nonlinear dynamics and chaotic theory				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Understanding of Nonlinear Dynamics	U	F	Internal Exam
CO2	Analyze the behavior of dynamical systems	Ap	P	Internal Exam
CO3	Exploration of Chaos Theory	U	C	Internal Exam

CO4	Numerical Analysis Skills	An	P	Internal Exam, Assignment
CO5	Apply the techniques of nonlinear dynamics to physical processes	Ap	P	Internal Exam, Assignment
CO6	Carry out simulation of Nonlinear systems	E	P	Assignment, Internal Exam
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge (F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs (48 +12)	Marks (70)
I	ONE DIMENSIONAL FLOWS		12	16
	1	Chaos, Fractals, Dynamics, Importance of Being Non linear	1	
	2	Flows on the line-Introduction, Geometric Way of thinking	1	
	3	Fixed points and Stability	2	
	4	Population Growth, Existence and Uniqueness	2	
	5	Saddle-Node, Transcritical, Pitch fork Bifurcations in One dimension	4	
	6	Imperfect Bifurcations and Catastrophes	2	
		Reference: Chapters 2, 3 of Book 1		
II	TWO DIMENSIONAL FLOWS		12	18

	7	Linear systems, Introduction, Definition and examples	2	
	8	Classification of linear systems	2	
	9	Phase Portraits, Fixed points and Linearization	2	
	10	Conservative systems	2	
	11	Reversible systems	1	
	12	Pendulum	3	
		Reference: Chapters 5, 6 of Book 1		
III	LIMIT CYCLES AND BIFURCATIONS		12	18
	13	Limit cycles-Introduction, Examples	2	
	14	Poincare-Bendixson Theorem	2	
	15	Relaxation Oscillators, Weakly Nonlinear Oscillations	2	
	16	Saddle-Node, Transcritical, Pitchfork and Hopf Bifurcations	2	
	17	Global Bifurcations of Cycles, Poincare maps	4	
		Reference: Chapters 7, 8 of Book 1		
IV	CHAOS		12	18
	18	Lorenz equations:- Introduction to Chaos and Properties of Lorenz equations	2	
	19	Chaos on a strange attractor, Lorenz map	2	
	20	One-dimensional Maps:- Fixed Points and Cobwebs	2	
	21	Logistic Map	2	
	22	Liapunov Exponent, Universality and Experiments	2	
		Reference: Chapters 9, 10 of Book 1		
V	OPEN ENDED MODULE: (ANALYTICAL AND/OR NUMERICAL TREATMENT OF NONLINEAR SYSTEMS)		12	

	FIXED POINT AND STABILITY OF ONE DIMENSIONAL SYSTEMS(ANALYTICAL) Phase portraits- two dimensional systems Numerical solutions of Simple pendulum Numerical study of Saddle-Node, Transcritical, Pitchfork and Hopf Bifurcations Numerical Integration of Lorenz systems Logistic map-Coweb Logistic map- Bifurcations, Liapunov exponent		
	Reference: 1. Nonlinear Dynamics and Chaos by S.H Strogatz (Book 1) 2. Nonlinear Dynamics: Integrability, Chaos and Pattersby M Lakshmanan & S Rajasekar 3. NPTEL video lectures: https://nptel.ac.in/courses/115106059		

Mapping of COs with PSOs and POs :

	PSO 1	PSO 2	PSO 3	PSO 4	PS O5	PS O6	PO1	PO2	PO3	PO4	PO5	PO 6	PO 7
CO 1	3	2	2	2	3	2	1	2	2	2	3	3	2
CO 2	3	2	3	2	3	3	3	2	3	2	3	3	1
CO 3	3	2	2	2	1	2	3	2	1	2	3	3	1
CO 4	3	2	3	2	3	3	2	2	3	2	3	3	2
CO 5	3	2	1	2	3	3	3	2	3	2	3	3	1
CO 6	3	2	2	2	3	3	3	2	2	2	3	3	2

Correlation Levels:

Level	Correlation
0	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Discussion / Seminar
- Internal Theory/Practical Exam
- Assignments /Viva
- End Semester Exam (70%)

Mapping of COs to Assessment Rubrics

	Internal Theory /Practical Exam	Assignmen t /Viva	Practical Skill Evaluation	End Semester Examinations
CO 1	✓	✓		✓
CO 2	✓	✓		✓
CO 3	✓	✓		✓
CO 4	✓	✓		✓
CO 5	✓	✓		✓
CO 6		✓	✓	